RRUHFR03

SDK Manual v1.5i

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About the Document

This document is applicable for RapidRadio UHF Integrated RFID Reader family products, i.e. RRUHFR03 and RRUHFR03R. It is intended to explain various functionality available in the SDK using detailed description along with application screenshots.

If you are looking for the User Manual or Communication Protocol Document, this is not the correct document. Please write back your actual requirements to us so we can provide you with the correct documentation.

After comparing this document with the application UI, a thought may arise that the document does not describe some of the functionalities available in the application. Please note that such configurations / operations are either not suitable or deprecated for new development and so the same have been omitted.

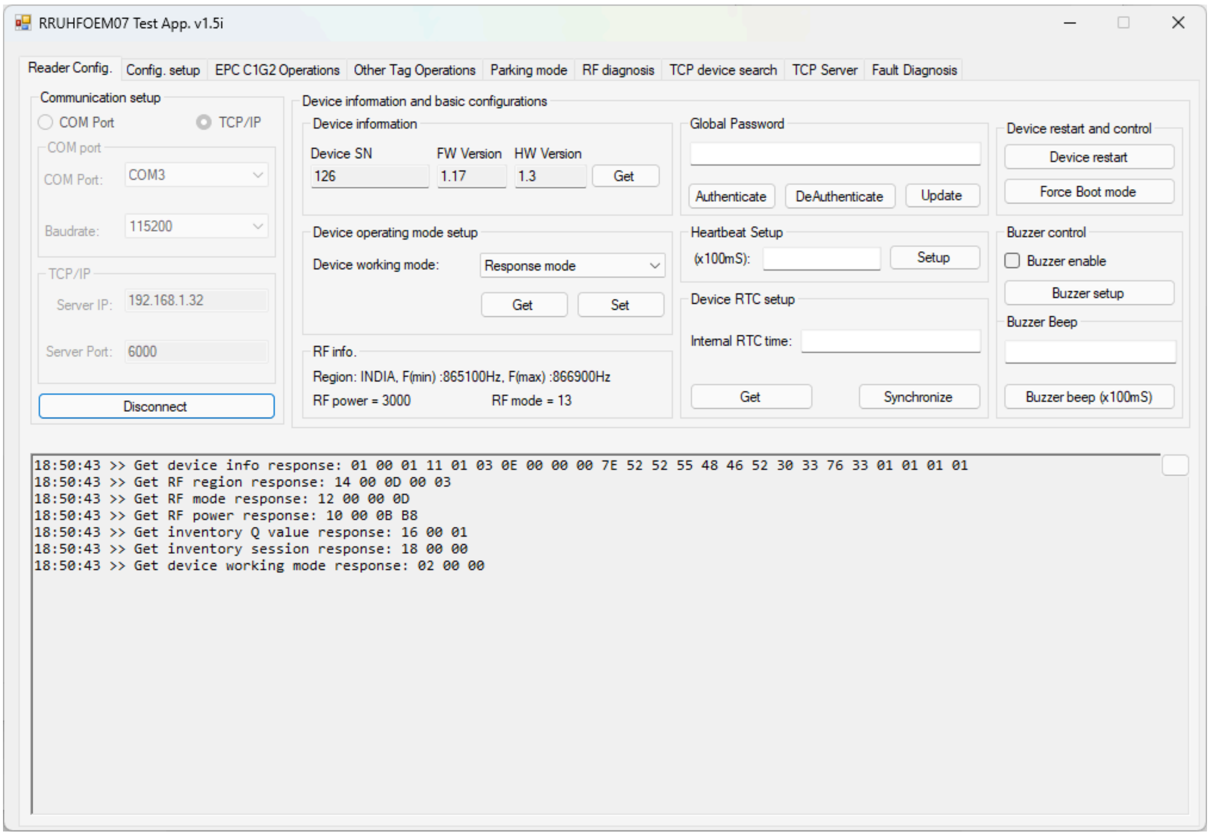
Tools Used

The SDK actually contains a Windows Forms Application based on .Net Framework 4.8, which has been coded in C#.Net. It uses bare metal code for demonstrating the RFID reader functionality, ensuring complete control of code to the system integrator. Only one external DLL named ZedGraph.dll is used to represent the RF noise data as a graph.

If you need the SDK for any other platform, feel free to write back to us. We may provide a basic SDK for the desired platform in case it is readily available with us. And even if it is not available, we can still help you integrate the RFID readers with your software.

Understanding the SDK

Reader Configurations



This screen contains basic configuration parameters for the RFID reader.

Communication Setup

The group contains a few configurations that are necessary for initiating a connection with the RFID reader. As it can be seen from the application, the application can connect to the RFID reader over a serial port or through LAN.

To connect the RFID reader using a serial port, please select the **COM Port** radio button and choose the correct port value along with the baudrate value from the list. In case you are unable to see any option to choose in the **COM Port** field, please close the application and execute the same once again after connecting a USB to serial converter cable. The default value of **Baudrate** is 115200.

For LAN connection, please choose the **TCP/IP** radio button and provide the **Server IP** (IP address assigned to the RFID reader) and **Server Port** (the listen port of the RFID reader). The default values of IP address and Port are 192.168.1.32 and 6000 respectively. In case the IP/port/both have been changed and you are unable to recall the values, you can use the **TCP Device Search** window.

Once all the information is provided in respective fields, press **Connect** button to initiate the connection with the RFID reader. After the connection is successful, the same button will be renamed with text **Disconnect**.

Device Information

In this group, basic information like device serial number, hardware version and firmware version can be fetched from the RFID reader by pressing the **Get** button. Device serial number is a 4 byte serial number that can be used to uniquely identify the device.

Device Operating Mode

The RFID reader supports a couple of operating modes, i.e. **Response Mode** and **Auto Mode**. Response mode is when the device will wait for any command received over any of the supported interfaces. Once a request is available, it will work on the command and respond back with the details. Whereas, when configured in the auto mode, the device will continuously scan for any new tags and as soon as a new tag is found, will push its EPC along with any other configured details to the remote server/host. **Get** and **Set** buttons can be used to either retrieve or configure the desired mode.

It is to be noted that to change any of the device configurations, the RFID reader is required to be configured in the Response mode first. It is made mandatory to ensure that the reader gives optimum performance during the Auto mode.

Heartbeat Setup

If enabled, the device can send heartbeat packets to the remote server/host at specified intervals. To enable the same, any non zero number shall be entered in the textbox before pressing the **Setup** button. The device will multiply the value with 100 and the resulting number will indicate the time difference between two consecutive heartbeat packets in milliseconds. The textbox will accept a 16 bit unsigned number for configuration of heartbeat, which is required to be provided in decimal format. The heartbeat can be enabled regardless of the work mode i.e. Auto mode or Response mode.

In case the RFID reader is configured in Auto mode, the reader will give first priority to the tag data packet. Heartbeat packet will be transmitted only if the configured time has elapsed after sending the previous heartbeat packet or the last data packet.

To disable the heartbeat functionality, press the Setup button with 0 value in the textbox.

Device Restart and Control

The button **Device Restart** can be used as a soft reset functionality to restart the RFID reader. Whereas the **Force Boot Mode** button will restart the RFID reader in bootloader mode so that an updated firmware can be programmed to the reader.

In case the reader is switched to bootloader mode by mistake, you can perform a power cycle to manually restart the same into the functional mode.

Buzzer Control

This group can be used to either enable or disable the automatic buzzer functionality. When enabled, the reader will beep for any successful operation. For Auto mode, the reader can indicate a tag detection event through beep sound when the buzzer is enabled. To disable the same, the **Buzzer Setup** button can be pressed after keeping the **Buzzer Enable** checkbox unchecked.

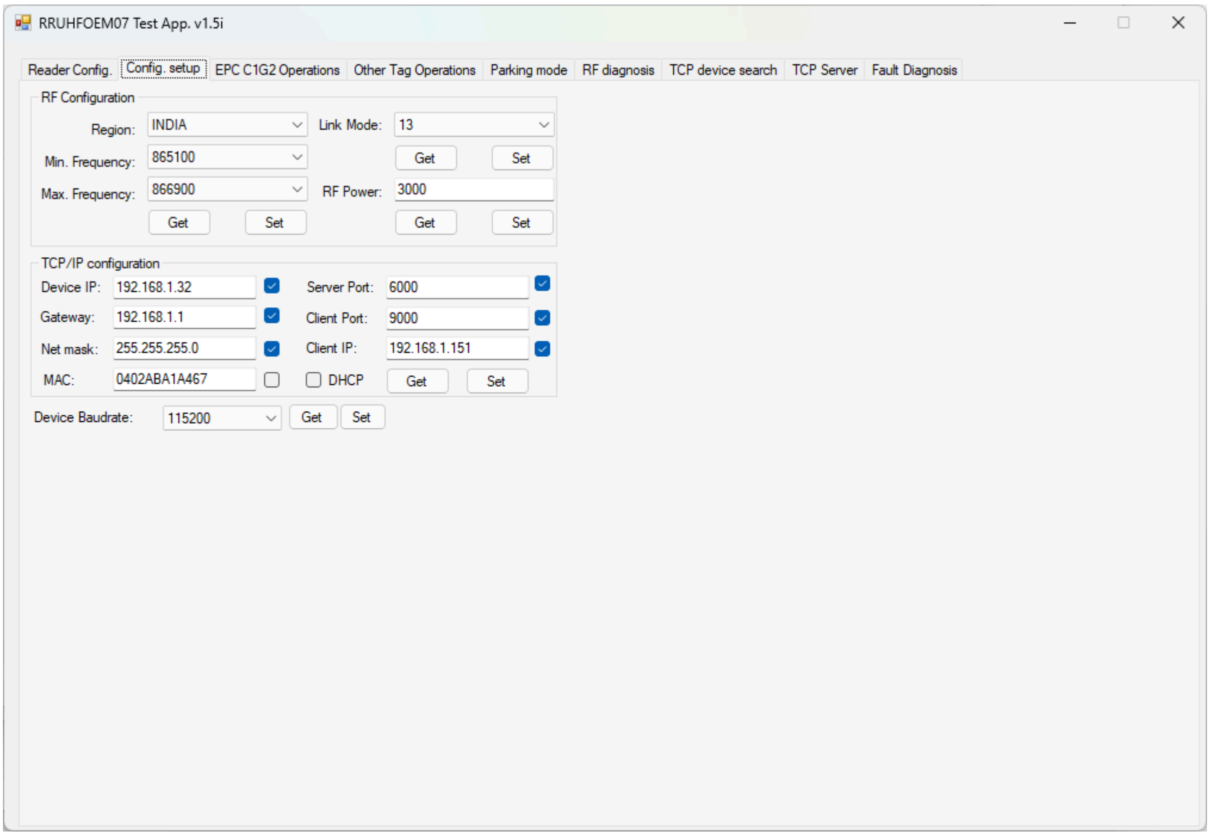
Buzzer Beep

This command can be used to force the RFID reader to give a beep sound manually. The duration of the beep can be provided by writing a 16 bit unsigned number as a decimal value in the textbox before pressing the **Buzzer Beep** button. This command will trigger beep sound even if the buzzer has been disabled using the **Buzzer Control** group.

Logs

Towards the bottom of the screen, a multiline textbox can be visible which has been used for logging. The textbox will show only the recent logs. The log window can be cleared using a small nameless button available at the right top corner of the textbox.

Configuration Setup



This screen contains additional configurations related to radio frequency and communication interfaces.

RF Configuration

This is a crucial configuration group and needs to be used with caution.

The **Region** contains a list of countries worldwide that have different frequencies allocated for UHF RFID devices. Depending on the country where the RFID reader is going to be used, the correct region should be chosen in order to avoid the violation of local policies. For India, one should select **INDIA** from the list, which is the default configuration.

Once the Region is changed, the **Minimum and Maximum Frequency** lists get updated. These configuration parameters can be used to limit the frequency hopping between the selected frequency range only. In case it is required to configure the device to communicate with the tags at a specific frequency (frequency hopping disabled), the same value should be used for both the fields, Minimum and Maximum Frequency.

To verify and configure the Region and Min & Max Frequency values, **Get** and **Set** buttons available below the fields.

Link Mode configuration can be used to configure the tag detection speed v/s performance. Some values may increase the tag detection speed but the read range will be reduced. Similarly, the configurations that can detect tags at longer distances will take longer time to complete an inventory cycle. The default value for Link Mode is 13 which is the optimum configuration for most of the applications. In case any specific application requires another value for better results, we strongly recommend to perform thorough on site tests with the new value before finalizing the same. The Get and Set button below this field can be used to check and change the value.

Reader performance depends on multiple parameters but the most important parameter out of the list is the **RF Power**. The reader supports output power of up to 33 dBm, which can be configured through software. However, it is not recommended to use power value above 30 dBm for longer time spans. The read range of the reader is proportional to the power value. The textbox accepts a four digit numeric value for power, out of which the right most two digits should be considered as decimal values. For eg. value 3000 means 30.00 dBm power. Similarly to set 25.5 dBm power, 2550 value should be configured in the textbox.

TCP/IP Configuration

This configuration group can be used to configure all parameters related to the ethernet communication.

Device IP is the IP address of the RFID reader and **Server Port** is the port on which the reader listens for incoming connection requests. Their default values are 192.168.1.32 and 6000 respectively. As these configurations are crucial to connect to the RFID reader in future, it is recommended to note them down after changing the same.

Once the device IP or port values are changed, a device restart is needed. Even though the Get button click provides the recently updated values, the changes take effect only after a hard reset or soft reset.

It is to be noted that the device does not support a public static IP address so it should be always configured with a suitable local IP address.

In case these configuration values are not known, **TCP Device Search** tab can be used to identify the same.

Gateway and **Subnet Mask** are required to communicate with different hosts connected in the network. They are configured with the default values 192.168.1.1 and 255.255.255.0 respectively.

Client IP and **Client Port** are basically the IP address and port of the remote server / host. These parameters will be useful when the reader is configured in Auto mode. If the configured IP address of a remote server / host points to a cloud instance, please make sure that the RFID reader is able to connect to the internet using the other TCP/IP configurations.

MAC Address of the RFID reader can be seen and if the need arises, can also be changed.

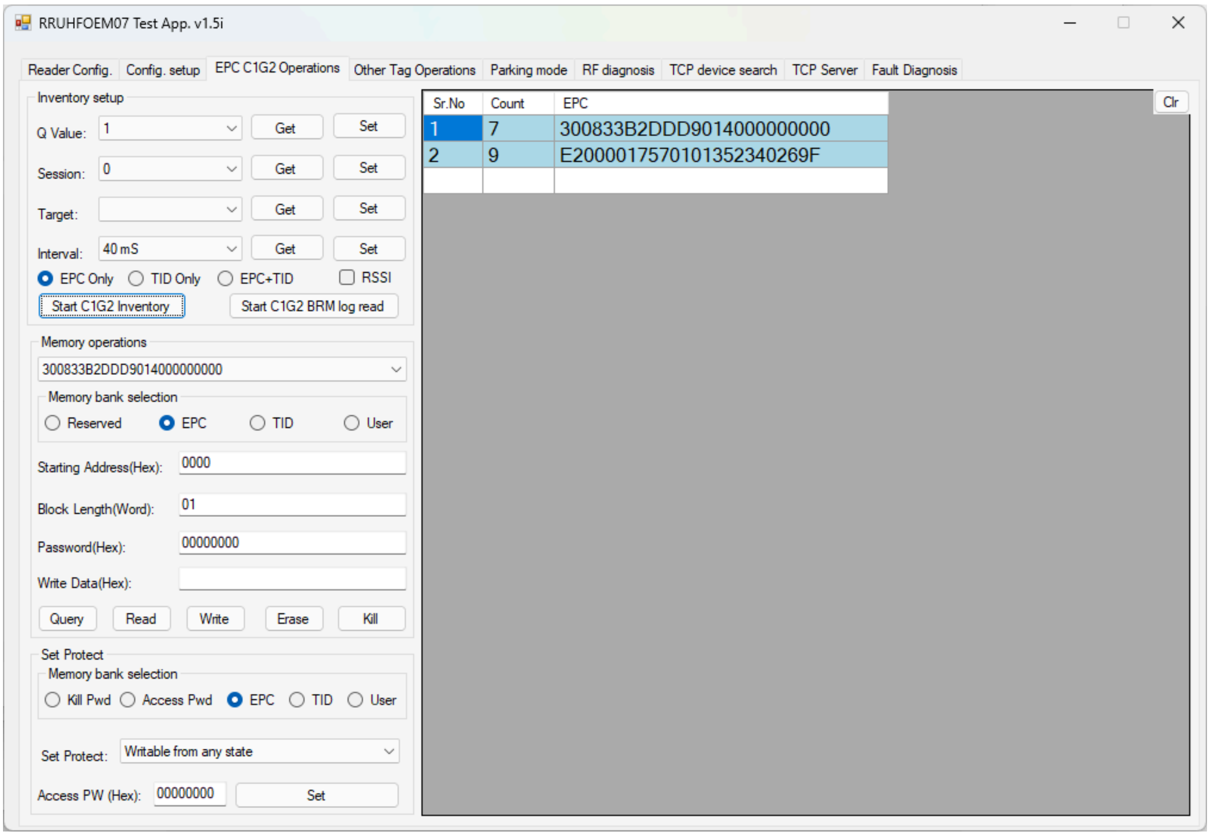
DHCP can be enabled in the RFID reader so that the device can fetch the IP address and related settings from a DHCP server. However, this can be useful only when the reader is configured in Auto mode.

To configure any of these values, the adjacent checkbox should be checked before pressing the **Set** button. As a precaution, the MAC checkbox is unchecked by default in order to prevent unintentional MAC address value changes. The **Get** button will read all configurations and show them on screen, irrespective of the checkbox status.

Device Baudrate

The baudrate of the serial communication can be changed between the provided values. The default value of baudrate is 115200.

EPC C1G2 Commands



As the name suggests, this screen has the commands related to ISO 18000-6C / EPC C1G2 standard. That means these commands can be executed only when the reader is configured in Response mode.

Inventory Setup

The group contains a few configurations related to the inventory operation.

Q Value, Session and Target are three parameters that, if jointly configured according to the need, can help quickly identify all the tags available near the RFID reader. All these configurations are applicable to both the operating modes i.e. Response mode and Auto mode.

Anytime, the total expected tags should always be in direct proportion to the square of **Q Value**. It's default value 4 means, up to 10 tags ($<2^4$ i.e. <16 tags) will be interrogated quickly by the RFID reader. Keeping a larger value for Q exponentially increases one inventory cycle time. Similarly for smaller than the required value of Q, too many collisions can increase the overall tag detection time. Thus choosing the correct value of Q as per the end application helps achieve faster detection speeds.

Using the **Session** value, the RFID reader can instruct tags to be on hold for some time after they have been interrogated by the RFID reader. This helps other tags, that are comparatively quieter (reflecting lower power), to get interrogated by the reader. It does not accurately define the timeline between two consecutive inventory responses, but still it is useful in quickly inventorying all the tags that are available near the RFID reader.

S0: No persistence after moving out from the RF field
S1: Persistence between 500ms and 5s regardless of whether tag stays or moves out of RF field
S2 and S3: Persistence of 2s or more after moving out from the RF field

Target is a slightly tricky configuration and thus the RFID reader handles it on its own. In case you need access to this parameter, feel free to write back to us.

Interval does not get stored inside the RFID reader, but rather it configures the SDK application to send the inventory command once after the configured time interval, when continuous inventory is enabled in the application by pressing the Start C1G2 Inventory button.

Start C1G2 Inventory button enables a timer for specified interval to continuously send the inventory command to the RFID reader. Do not mix up this operation with the Auto mode of the RFID reader. In Auto mode, the reader keeps on looking for new tags on its own, whereas during this operation, the application keeps sending an inventory command to the RFID reader at the configured time interval. It is obvious that the Auto mode is the fastest possible way to detect tags quickly. However, specific applications may need the RFID reader in Response mode so to help them integrate the reader faster, we implemented the continuous inventory mechanism.

Additionally, a few radio buttons and a check box are available over this button. **EPC Only**, **TID Only** and **EPC+TID** radio buttons instruct the reader to send specific data of the tag when responding to the inventory command. If **RSSI** is checked, RSSI value will also be added in the inventory response frame.

Memory Operations

Tag memory read, write and erase operations are added in this group. It also contains a button to kill a tag. Before performing any memory related operation over a tag, it is mandatory to perform inventory operation. If the inventory completes successfully, the EPC value of all the detected tags will be stored in the combo box for convenience. User needs to select the desired tag from the list and proceed further to perform the memory operation.

The **Memory Bank Selection** group has four radio buttons indicating various memory sections available in a UHF tag. The reserved memory bank stores the passwords (4 byte Kill password in blocks 0 & 1 and 4 byte Access password blocks 2 & 3) whereas the EPC, TID and User sections are dedicated to store the concerned data.

All the sections of tag memory are divided into a fixed number of blocks, depending on the tag chip. Each such block contains 2 bytes of memory.

Starting Address is a 16 bit unsigned integer, to be written as four hex characters in MSB first order. It denotes the starting block number of the selected memory area, where the operation is required to be performed. Please note that to read/write EPC memory, the starting address should always be 0002 as the first two blocks of the EPC memory bank store some other information. For all other memory banks, the value can start with 0000.

It is to be noted that the application required four hex digits to perform this operation. If the value 0002 is provided as a single byte 02, the application won't be able to send the command to the RFID reader.

Block Length indicates total blocks that are required to be read/written. As each block has 2 bytes of memory, block length of 6 means the operation needs to be performed over 12 bytes of data. Being an 8 bit unsigned integer, this value shall be written as two hex characters.

Password is basically the access password that has been pre-configured in a specific tag to protect the tag memory from unauthorized access. If a specific memory section of a tag is protected, a 4 byte access password is required to be provided to the RFID reader so the tag memory can be authenticated before performing the operation. The default access password value is 00000000 (as a hex string).

Write Data textbox is used to provide data to be written in the tag memory. Similarly for read operations, the data read from tag memory will be shown in the same textbox.

By pressing the **Query** button, the inventory command can be sent to the RFID reader. As described earlier, it is mandatory to do inventory before performing read/write operations over the tag memory. Unlike the Start C1G2 Inventory button, this button will perform inventory operation only once.

Read, **Write** and **Erase** buttons can be used to perform specific operations over the tag memory. The erase operation is actually the same as the write operation, as it writes 0x00 in the tag memory.

The **Kill** button is intended to kill the selected tag so that it does not respond to any of the commands in future. This command requires the kill password to be transmitted along with the tag EPC value. Only if the tag's kill password is not the default value (00000000) and if it matches with the provided value of kill password, the tag will be killed. The kill operation is irreversible, so please use it with caution.

Set Protect

Just changing the access password does not protect a UHF tag for read/write operations. Various memory blocks are required to be protected individually in order to do so.

To protect a specific memory section of a UHF tag, a respective radio button from the **Memory Bank Selection** group is required to be checked. Along with that, the protection level can be selected in the **Set Protect** field. Additionally, the **Access Password** is also required to be provided so the tag memory can be protected. After entering all the required details, pressing the Set button shall set the protection.

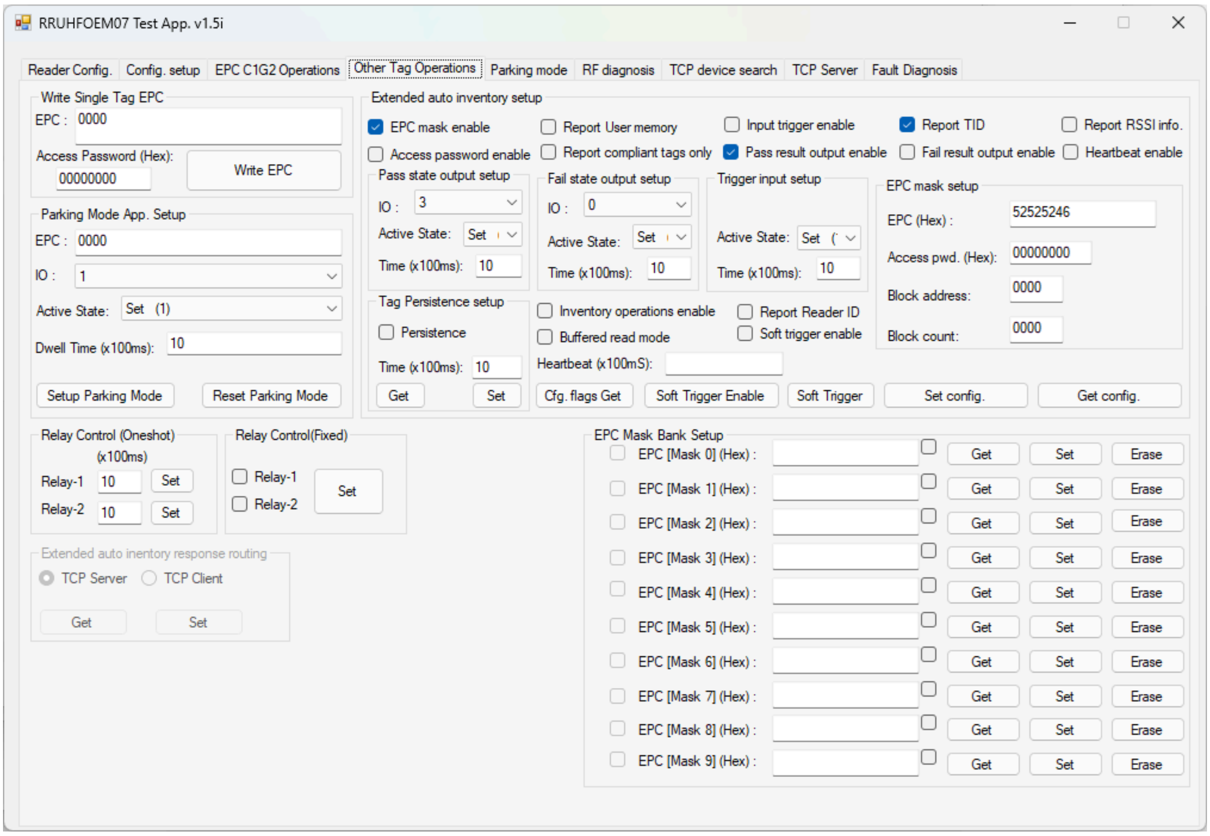
The process is required to be repeated for all memory banks, in order to protect the entire tag.

Tag List

On the right side of the screen, a list is visible. Whenever a tag is detected, either by pressing the Query button or using Start C1G2 Inventory functionality, the tag will be listed here. If an entry is already there in the list for a specific tag, the detection count will increase. Depending on the selected radio button in the Inventory Setup group, necessary columns will be automatically added in/removed from the list.

The list can be cleared using a small **Clr** button available on the top right corner of the list.

Other Tag Operations



Other tag operations window includes a few EPC C1G2 commands and additional configurations related to the Auto mode of the RFID reader.

Write Single Tag EPC

By default, almost all the UHF tags have a 12 byte long EPC. However, the EPC length can be reduced and for some tags, it can be even extended. To do so, one tag is required to be kept near the reader and before pressing the **Write EPC** button, the desired EPC value and the access password are required to be entered in respective textboxes.

It is to be noted that there should be only one tag available near the reader when performing this operation.

Relay Control (Oneshot)

This command can be helpful in setting the desired relay for a specific time interval. The textboxes accept a 16 bit unsigned integer value which will be multiplied by the reader with 100 in order to derive total milliseconds for which the relay will stay active. After that time interval, it will be automatically switched off.

Relay Control (Fixed)

Using this command, one can operate multiple relays at once. However, the same won't be disabled automatically. Instead a separate command is required to be sent to the reader to switch the relays off. To set the relays, the checkbox adjacent to the relay that is required to be operated needs to be checked before pressing the **Set** button. The relays associated with unchecked checkboxes will be disabled.

Extended Auto Inventory Setup

The RFID reader has a few flags to select from when it is being configured in Auto mode. Depending on these flag values, the RFID reader will include required information in the auto inventory tag data frame.

When **EPC Mask Enable** is checked, the RFID reader will internally filter out the tags based on their EPC value. Only if the tag's EPC value starts with the data written in the **EPC** field available inside the **EPC Mask Setup** group, the tag will be considered a valid tag. Rest all tags will become invalid tags. Based on such validity status of a tag, further process can be initiated by the RFID reader.

Report User Memory can be checked in order to inform the RFID reader to submit a specific portion of the tag user memory along with the EPC value. In case the tag does not contain a user memory section in its memory, the reader will still send the data of that tag without user memory contents. **Block Address** and **Block Count** fields are used to configure the portion of user memory. Block Address is 16 bit unsigned value whereas block count is 8 bit unsigned value.

On checking **Report TID**, the TID value of a tag can be requested in the auto inventory tag data frame. Similarly, if RSSI is also desired in the tag data frame, **Report RSSI** can also be checked.

Similar to the EPC Mask Enable flag, the **Access Password Enable** flag can also be used to filter tags. The checkbox can be ticked to filter all the tags that are pre-configured with a specific Access Password value. The value of access password shall be provided in the **Access Pwd** field of EPC Mask Setup group.

When both the options EPC Mask Enable and Access Password Enable are checked, they both will jointly decide if a tag is valid or not. A tag matching both the criteria will be considered a valid tag.

By selecting **Report Compliant Tags Only**, the RFID reader can be instructed to send only the valid tag data frames. If it is unchecked, data of all the tags, valid and invalid, will be sent to the application.

Input Trigger Enable checkbox can be used to configure a hardware trigger functionality. Further configurations related to this are available in the **Trigger Input**

Setup group. To set up a trigger, **Active State** shall always have **Reset** value. In addition, the time interval for which the inventory operations shall be active after an external trigger, can be configured using the textbox available to write a 16 bit unsigned number. As the number is multiplied with 100 internally by the reader, value 10 means the auto inventory operation will be active for 1s after the input trigger.

The **Pass Result Output Enable** field can inform the RFID reader to automatically operate a specific relay as soon as a valid tag is found. The additional configurations of relays can be done in the groupbox **Pass State Output Setup**. To operate relay 1, IO 3 is required to be selected and for relay 2, IO 2 shall be selected. Active State shall always be Set (1). In addition, the relay operation time can be written in the adjacent textbox, which will accept a 16 bit unsigned value. Value 1 will trigger the relay for 100ms whereas 10 will set the relay for 1s.

Similarly, if checked, **Fail Result Output Enable** will trigger a relay for invalid tags. For the configurations, **Fail State Output Setup** group can be used.

Please note that once any relay has been used for either valid tags or for invalid tags, the same relay cannot be operated using discrete commands.

The **Heartbeat Enable** checkbox is just another shortcut from where you can enable the heartbeat functionality. You can check the detailed description of heartbeat in the Reader Configurations section. If this checkbox is checked, the 16 bit unsigned value of heartbeat interval can be configured using the Heartbeat textbox available towards the bottom of the group.

Tag Persistence Setup group can be used to forcefully ask the RFID reader to prevent repeating the data packet of the same tag for a certain period of time. To enable the persistence, the checkbox is required to be checked and the 16 bit unsigned value of persistence time shall be configured in the textbox.

When persistence is enabled, the RFID reader will check the time difference between two consecutive detections of the same tag and compare it with the time configured as persistence time. If the time difference is higher than the persistence time, the tag data will be sent to the remote server/host. Otherwise the tag detection will be omitted.

When enabled, the **Report Reader Id** flag will include the 4 byte reader serial number in the tag data packet.

The checkbox **Inventory Operations Enable** is required to be checked/unchecked to respectively enable/disable the auto inventory operation. In other words, the RFID reader considers itself in Response mode when the checkbox is unchecked. And when checked, it represents the reader in Auto mode.

If an RFID reader is going to be used in Auto mode and for some reason, it is required to be configured in Response mode temporarily, it is recommended to change its mode using the Device Working Mode parameter, available on Reader Reader Configurations screen. This will ensure that all other parameters related to Auto mode are intact. So later, when the mode is switched back to the Auto mode, there is no need to set all the configurations again.

Soft Trigger Enable checkbox and button, both can enable software trigger functionality. That means by using this functionality, when the **Soft Trigger** button is pressed, the RFID reader performs the inventory process once and sends the data of detected tags to the remote host.

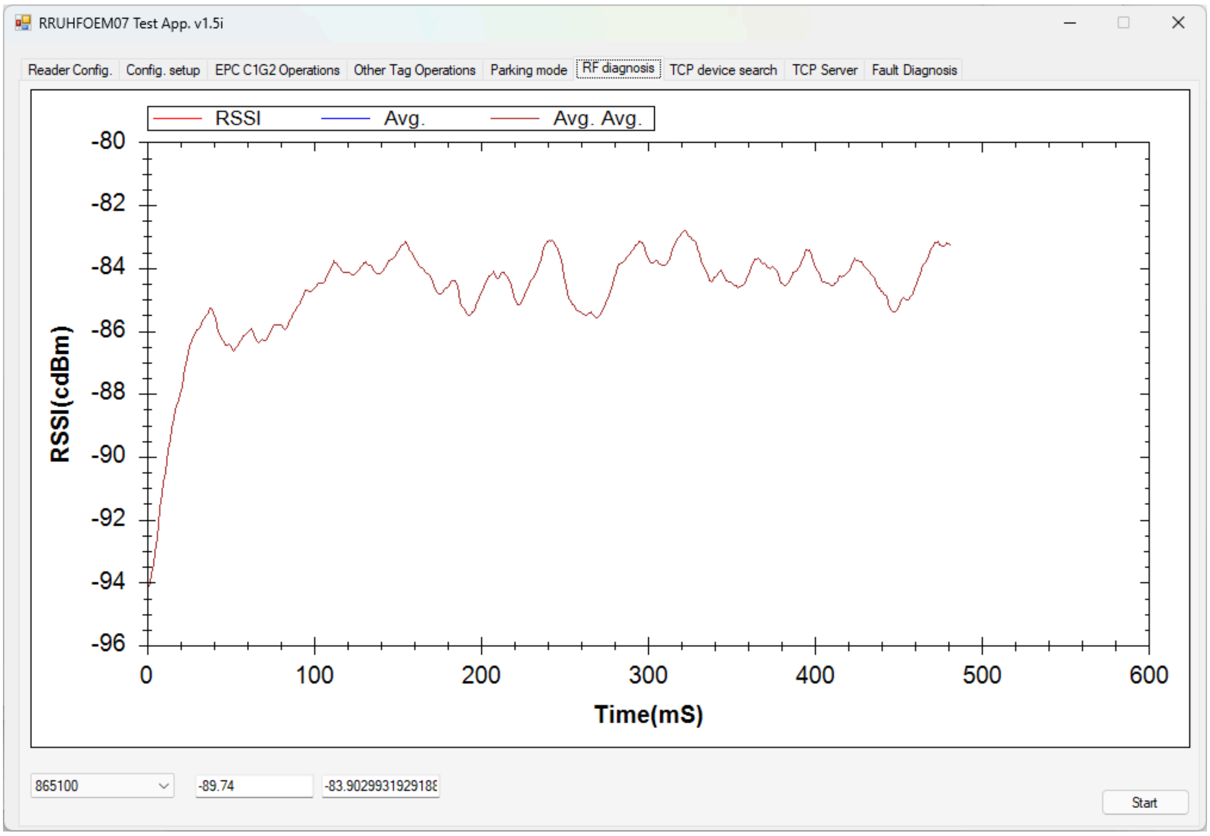
All the above configurations available under Extended Auto Inventory Setup can be saved and retrieved using **Set Config** and **Get Config** buttons. Even if some of the checkboxes are changed in the application, the RFID reader will not be able to know the same unless the same is configured inside the reader by pressing the Set Config button.

EPC Mask Bank Setup

This group contains controls to configure multiple EPC masks in the RFID reader, so that if a tag EPC value matches with any of the configured mask value, the tag will be considered a valid tag. This can be helpful when the details of tags having different EPC initial values are required to be passed on to the software. It is possible to configure a total 10 different EPC masks into the RFID reader. The first mask value is shared between this group and the EPC Mask Setup group.

To add a mask, please make sure you get details of all mask settings starting from first to the last one, by pressing the respective **Get** button. Once you find a blank EPC mask setup, write the desired mask value in the textbox, enable the checkbox associated with the mask setup and press the **Set** button to save the same in the RFID reader. To disable a mask setup, the checkbox should be unchecked before pressing the Set button. If the record is required to be deleted entirely, the **Erase** may come to the help.

RF Diagnosis

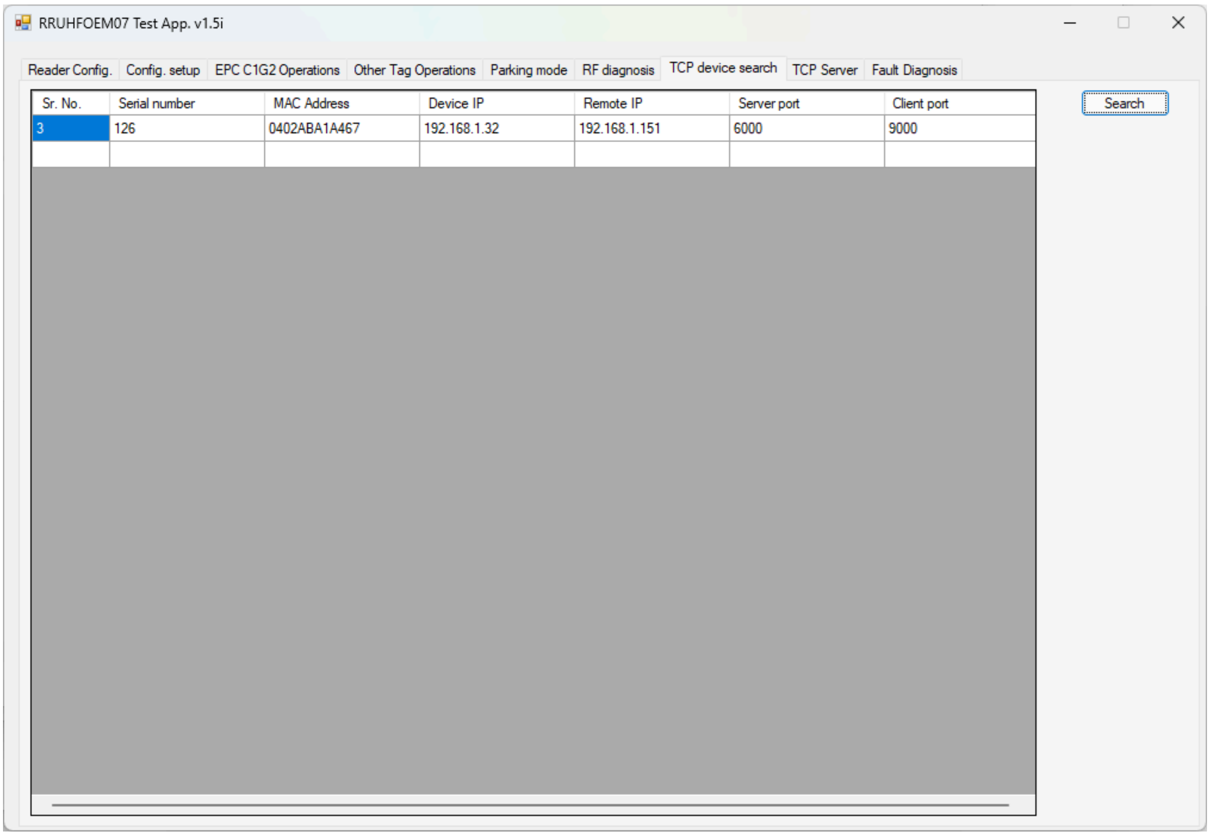


The application screen helps to check the environmental noise level. It can greatly help in troubleshooting the on site performance issues.

To start testing, one has to select a specific band from the left bottom side list and press the Start button. The data received by the application will be arranged in a graph for better understanding.

At any point of time, the RSSI value shown on the graph should be less than -80 dBm for optimum performance of the RFID reader. For RSSI values between -70 dBm and -80 dBm, a very minor performance lag may or may not be observed, depending on the other parameters. In case of RSSI values higher than -70 dBm (towards 0 dBm), it is possible that visible performance difference is observed.

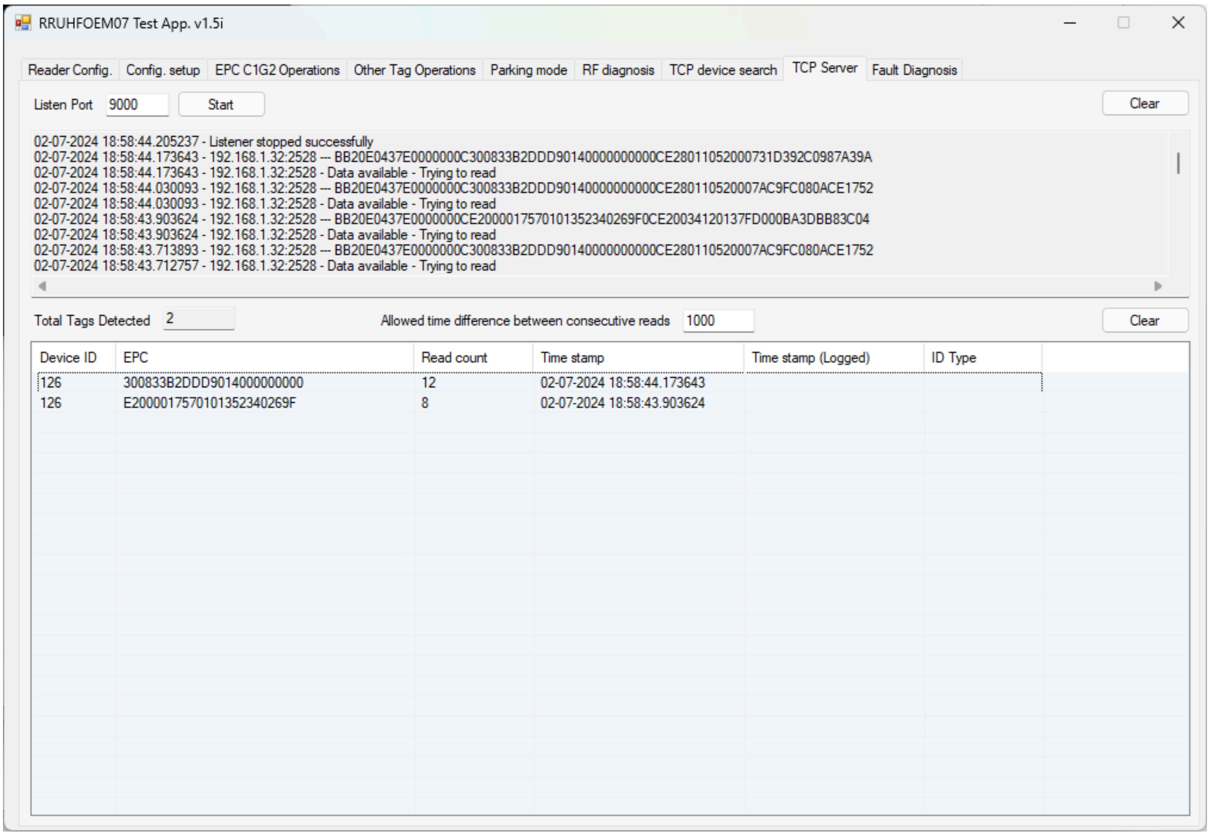
TCP Device Search



If the IP address or port of an RFID reader has been changed and the user is unable to recall the same, this tab can be helpful to find the details. If the **Search** button is pressed, the application will list all the RFID readers available in the network along with their IP address, port and a few other information.

If you are unable to list the RFID readers in the application, please may like to check after disabling the firewall.

TCP Server



This tab implements a generic TCP server that listens on the chosen port and logs all the packets received on that specific port in the textbox. In addition, if the received packet happens to be a tag data packet from the RFID reader, the contents are shown in the available list. Both the textbox and the list can be cleared anytime by pressing the respective Clear button.

In case you have configured a TCP listener on a specific port in your application, the application will not be able to listen on the same port. Similarly, you won't be able to listen on the same port unless you stop the listening in this application.

It is worth noting that the tag data frames received by the TCP server will also be added in the list available at EPC C1G2 Commands screen.

Revision History

Sr. No.	Revision Date	Details	Revised By	Approved By
1	08/07/2024	Initial Draft	DSP	DCU